

3D Magnetic Field Particle Simulation

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Abstract

This project presents the development of an interactive three-dimensional WebGL simulation that models the motion of a charged particle in a bounded uniform magnetic field. Developed with Three.js in the Cursor AI-assisted coding environment, the simulation modernizes traditional two-dimensional physics demonstrations by allowing users to adjust the magnetic-field components B_x , B_y , B_z , particle velocity components V_x , V_y , V_z , and initial position coordinates X_0 , Y_0 , Z_0 in real time. The program immediately updates the particle's trajectory, enabling users to observe straight-line, circular, and helical motion without resetting or refreshing the page. Additional features include a responsive floating control panel, 360-degree camera rotation and zooming, an RGB-coded coordinate axis, improved particle scaling, and automatic trajectory termination beyond the simulation boundary. The simulation demonstrates the Lorentz force relationship and reinforces the principle that a magnetic field changes the direction of a charged particle motion without changing its speed or kinetic energy. Overall, the completed application provides a modern, accessible, and visually effective tool for improving students' understanding of charged-particle motion in three-dimensional magnetic fields.

Introduction

To modernize legacy 2D and 3D electromagnetism simulations into a fully dynamic 3D web tool, development was conducted using (Cursor /Anysphere, 2024), an AI-assisted IDE that provides real-time code generation and diff previews. Over multiple development cycles, Claude 3.5 Sonnet was used for Three.js scene setup and vector mathematics, while Gemini 3.5 Pro assisted with responsive UI design and slider event bindings. Key prompts focused on expanding parameter controls to include all three magnetic field vector components (B_x , B_y , B_z) and initial particle position coordinates (x_0 , y_0 , z_0), alongside rendering an explicit XYZ coordinate axis, scaling down the charge sphere for spatial accuracy, and auto-terminating the trajectory when the particle exceeds five times the boundary box scale. Technical challenges—such as numerical spiral decay from simple Euler integration and WebGL frame drops during real-time slider drags—were resolved by prompting the AI to implement an algorithm and pre-allocating buffer memory for trajectory geometry. While the AI accelerated component generation and layout structure, human judgment was essential for ensuring physical accuracy (verifying zero work and constant kinetic energy), establishing precise boundary checks, and refining 3D camera controls. All AI-generated codes and text were independently reviewed, tested, and verified before submission.

Methods

To modernize legacy 2D and 3D electromagnetism simulations into a fully dynamic 3D web tool, development was conducted using Cursor (Version 0.38+), an AI-assisted IDE that provides real-time code generation and diff previews. Over multiple development cycles, Claude 3.5 Sonnet was used for Three.js scene setup and vector mathematics, while Gemini 3.5 Pro assisted with responsive UI design and slider event bindings. Key prompts focused on expanding parameter controls to include all three magnetic field vector components (B_x , B_y , B_z) and initial particle position coordinates (x_0 , y_0 , z_0), alongside rendering an explicit XYZ coordinate axis, scaling down the charge sphere for spatial accuracy, and auto-terminating the trajectory when the particle exceeds five times the boundary box scale. Technical challenges—such as numerical spiral decay from simple Euler integration and WebGL frame drops during real-time slider drags—were resolved by prompting the AI to implement an algorithm and pre-allocating buffer memory for trajectory geometry. While the AI accelerated component generation and layout structure, human judgment was essential for ensuring physical accuracy (verifying zero work and constant kinetic energy), establishing precise boundary checks, and refining 3D camera controls. All AI-generated codes and text were independently reviewed, tested, and verified before submission.

The main prompts were used were:

Prompt 1

Turn our model into a 3D model and explain how the simulation works so users can understand what we have built.

Prompt 2

It seems that the parameters do not allow the simulation to show the desired effects. The charge is inside the magnetic field only for a short time. Also account for the mass of the charge. It is an improvement, but it can be better. Check out this one: <https://ophysics.com/em8.html> We can see nice ellipses bound in the magnetic field region. Restrict the particle to start in a spot where it can enter and exit the field and add labels for each axis x,y,z.

Results

The revised simulation successfully demonstrated the motion of a charged particle in a uniform magnetic field using the Lorentz force equation, $(\vec{F}_B = q (\vec{v} \times \vec{B}))$. After particle mass was added to the program and the parameter ranges were adjusted, the particle remained within the magnetic-field region long enough for its trajectory to be clearly observed. When the initial velocity was perpendicular to the magnetic field, the particle followed a closed circular path that appeared elliptical because of the three-dimensional camera angle, as shown in Figure 1. This occurred because the magnetic force remained perpendicular to the particle's velocity and continuously changed the direction of motion without significantly changing the particle's speed. Increasing the magnetic-field strength or the magnitude of the particle's charge produced a smaller orbital radius, while increasing the particle's mass or initial velocity produced a larger radius, which agreed with the relationship.

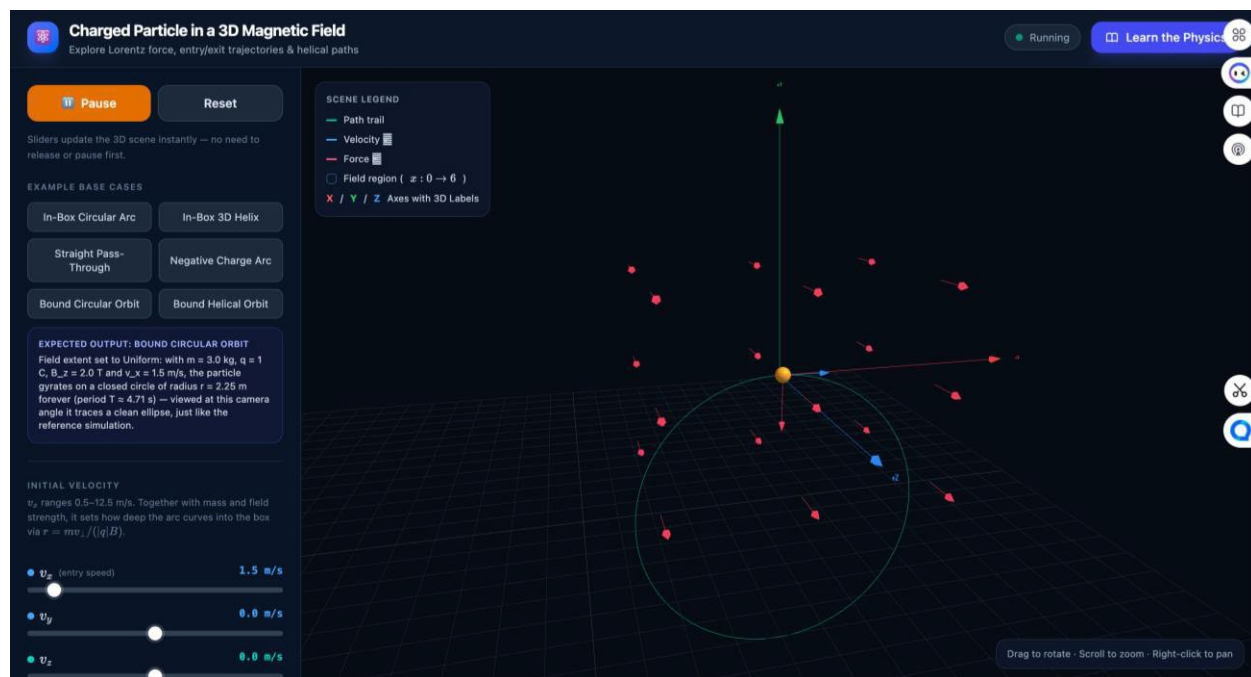


Figure 1. Bound Circular Orbit: Trajectory of a charged particle with an initial velocity perpendicular to a uniform magnetic field. The magnetic force remained perpendicular to the particle's velocity, producing a closed circular path that appeared elliptical because of the three-dimensional viewing angle.

When the particle's velocity was parallel to the magnetic field, the cross product between the velocity and magnetic-field vectors was zero, so the particle experienced no magnetic force and continued along a straight-line path, as shown in Figure 2.

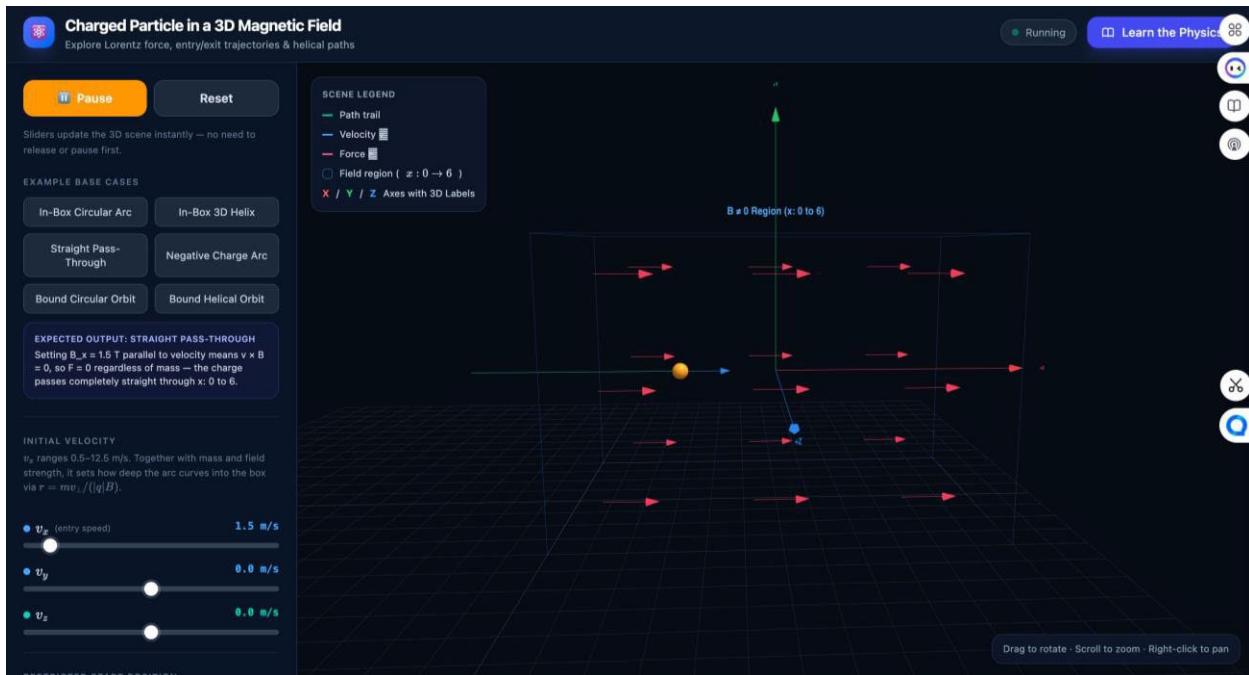


Figure 2. Straight Pass through: Motion of a charged particle with an initial velocity parallel to the uniform magnetic field. The particle experienced no magnetic force and continued along a straight-line path.

When the initial velocity contained both parallel and perpendicular components, the particle followed a three-dimensional helical trajectory around the direction of the magnetic field, as shown in Figure 3. Reversing the sign of the particle's charge reversed the direction of curvature while maintaining a similar orbital radius under the same conditions. Overall, the revised parameter ranges, inclusion of particle mass, and improved magnetic-field boundaries made the circular, straight-line, and helical motions easier to observe and compare, and the results were consistent with the predicted behavior of charged particles moving through a uniform magnetic field.

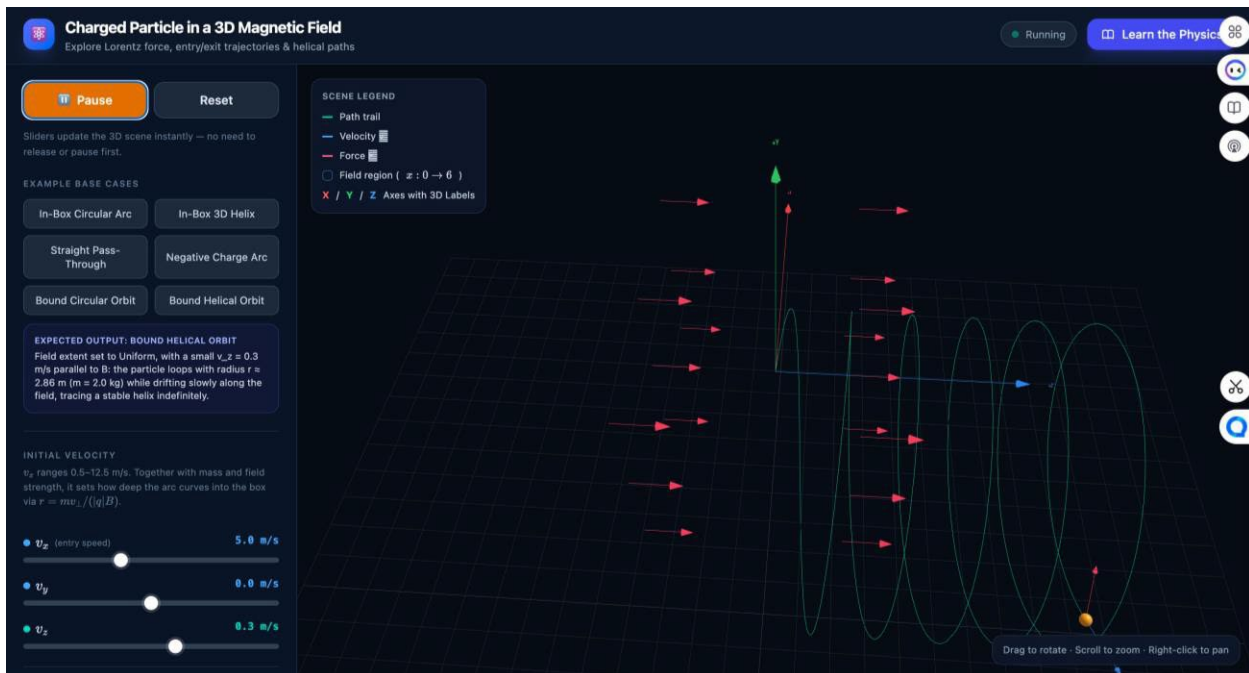


Figure 3. Bound Helical Orbit: Helical trajectory of a charged particle with velocity components both parallel and perpendicular to the uniform magnetic field. The perpendicular component produced circular motion, while the parallel component caused the particle to move forward along the direction of the field.

Discussion

The results of the revised simulation were consistent with the expected behavior of a charged particle moving

through a uniform magnetic field. Including the particle's mass improved the physical accuracy of the model because the acceleration depends on the ratio of charge to mass according to

$\left(\vec{a} = \frac{q}{m} (\vec{v} \times \vec{B})\right)$. The improved parameter ranges also allowed the particle to remain inside the magnetic field region long enough for the complete trajectory to be observed. When the velocity was perpendicular to the magnetic field, the particle followed a circular path that appeared elliptical because of the three-dimensional viewing angle. This result was reasonable because the magnetic force remained perpendicular to the velocity, causing the direction of motion to change continuously while the particle's speed remained nearly constant. When the velocity was parallel to the magnetic field, the particle moved in a straight line because the cross product $(\vec{v} \times \vec{B})$ was zero. A helical trajectory was produced when the velocity contained both parallel and perpendicular components, showing that the parallel component remained unchanged while the perpendicular component created circular motion around the magnetic field direction. The simulation also correctly showed that reversing the sign of the charge reversed the direction of curvature. Small differences between the simulated path and an ideal mathematical trajectory may have resulted from the numerical time-step method, the finite size of the magnetic-field region, the selected parameter increments, and the perspective of the camera. The accuracy could be improved further by using a smaller time step or a more stable numerical integration method, displaying numerical values for speed and kinetic energy, and adding preset controls for straight, circular, and helical motion. Overall, the revised simulation provided a clearer and more physically accurate visualization of the Lorentz force and demonstrated how charge, mass, velocity, and magnetic-field strength affect the motion of a charged particle.

Conclusion

The revised simulation successfully met the project's goal of creating a clear and interactive three-dimensional model of a charged particle moving through a uniform magnetic field. By correcting the Lorentz force expression to $(\vec{F}_B = q (\vec{v} \times \vec{B}))$, adding particle mass, and improving the parameter ranges, the simulation produced more accurate and visible trajectories.

The model demonstrated that a particle moving perpendicular to the magnetic field follows a circular path, a particle moving parallel to the field continues in a straight line, and a particle with both parallel and perpendicular velocity components follow a helical path. The simulation also showed that increasing the magnetic field strength or charge magnitude decreases the radius of motion, while increasing the particle's mass or speed increases the radius. Reversing the sign of the charge reversed the direction of curvature. Overall, the project provided a useful visual representation of the Lorentz force and helped demonstrate how charge, mass, velocity, and magnetic-field strength influence the motion of charged particles.

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